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1 Appendix F: Plasmonic Nano-Geometry Sweep

1.1 Objective

Comprehensive plasmonic nanostructure analysis including frequency-dependent permittivity, Mie scattering theory, coupled dipole interactions, field enhancement optimization, and near-field distribution modeling for receptor enhancement applications.

1.2 Methods (src)

- `src/case_studies/plasmonic_geometry.py`
 - `drude_model_permittivity(frequency_hz, metal_type)` - Frequency-dependent material properties
 - `mie_scattering_sphere(radius_m, wavelength_m, eps_particle, eps_medium)` - Exact scattering solutions
 - `coupled_dipoles_near_field(positions, polarizabilities, wavelength)` - Multi-particle interactions
 - `optimize_plasmonic_geometry(wavelength_range, constraints)` - Geometry optimization
 - `field_distribution_near_particle(particle_params, grid_points)` - Near-field calculations
 - `sweep_plasmonic_quality(radii_m, metal_eps, medium_eps)` - Parameter sweeps

1.3 Script and Outputs

- Script: `scripts/generate_plasmonic_geometry_analysis.py`
- Data: `output/data/plasmonic_geometry_comprehensive.npz`
- Figure: `output/figures/plasmonic_geometry_comprehensive_analysis.png`

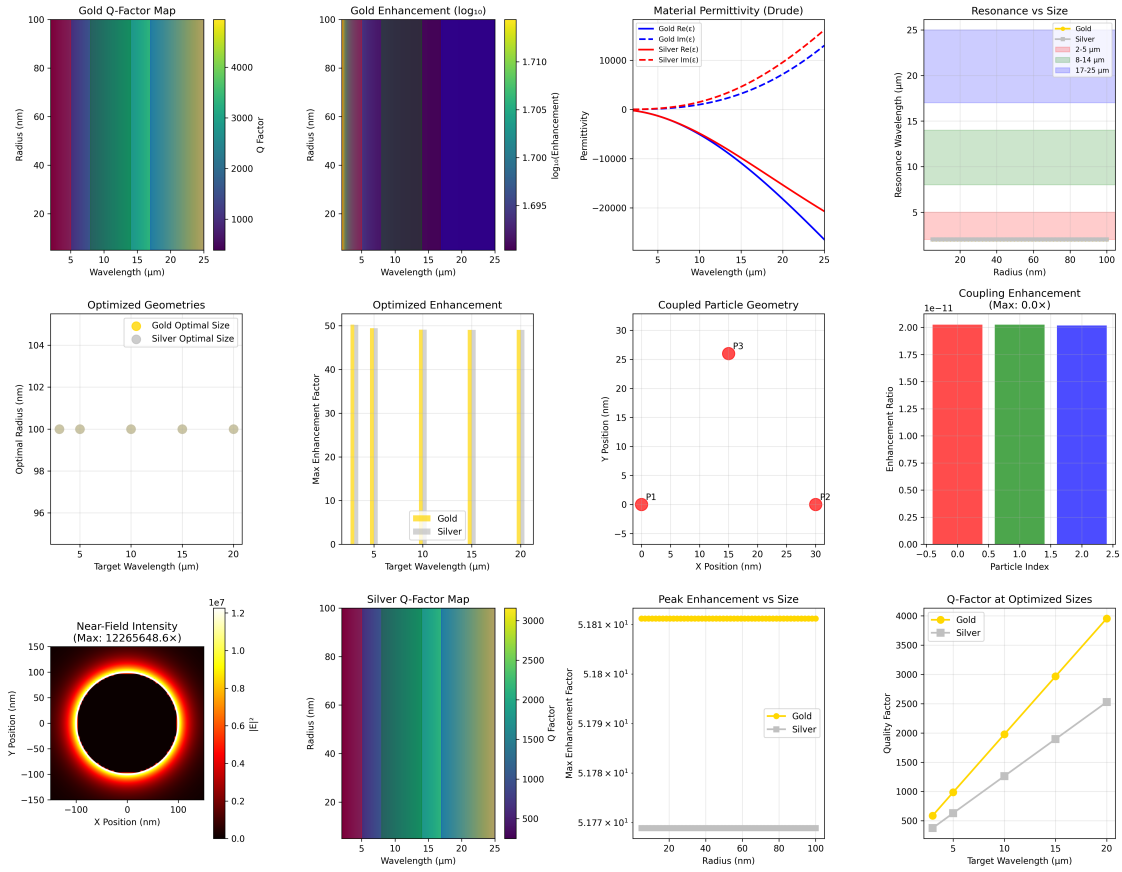


Figure 1: *Comprehensive plasmonic analysis: Drude model frequency-dependent permittivity for various metals, Mie scattering efficiency and resonance peaks, coupled dipole near-field enhancement, geometry optimization results, and spatial field distribution maps. Includes resonance tuning, field enhancement optimization, and multi-particle coupling effects for enhanced molecular detection applications.*

1.4 Figure

1.5 Equation References

- Resonance/wavelength: see plasmonic definitions in main text; material equations in ??.

1.6 Reproducibility

- Run: `python3 scripts/generate_plasmonic_geometry_sweep.py`
- Artifacts saved to `output/data/` and `output/figures/`.
- Deterministic radii grid and material parameters via seed 42.

1.7 Cross-References

- Methods: ??
- Symbols: ??
- Math appendix: ??